Enzyme kinetics

Enzyme kinetics is the study of the rates of **enzyme-catalyzed reactions** and how they are affected by different factors.

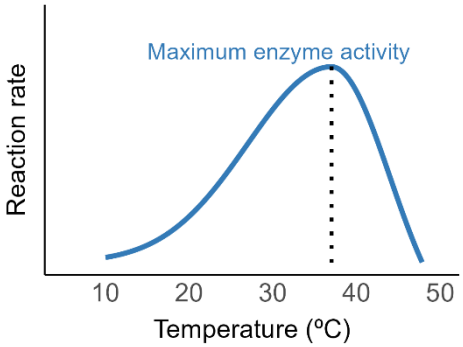
The study of the kinetics of an enzyme can reveal its catalytic mechanism, its role, its regulation or how its activity can be affected by an inhibitor or a drug.

The rate of an enzyme-catalyzed reaction can be affected by:

- Substrate concentration
- Enzyme concentration
- Cofactors and regulators
- Temperature
- pH

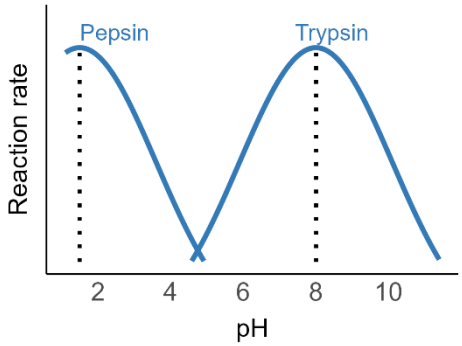
Effect of temperature on the rate of enzyme-catalyzed reactions

As with many chemical reactions, **the rate of an enzyme-catalyzed reaction increases as the temperature increases**. Variations as small as 1 or 2 degrees may introduce changes of 10 to 20% in the reaction rate. **The optimum temperature for most human enzymes is around 37ºC**. At higher temperatures the rate decreases because, among other factors, the enzyme becomes denatured.



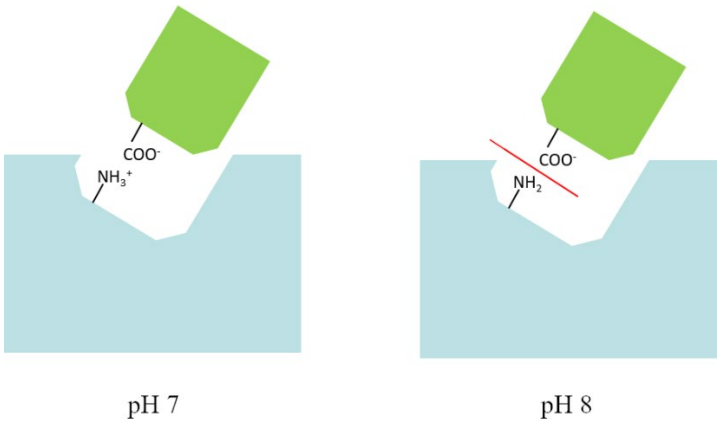
Effect of pH on the rate of enzyme-catalyzed reactions

The optimum pH for an enzyme is generally close to the pH of the environment in which the enzyme is found.



Pepsin works in the highly acidic conditions of the stomach. Accordingly, pepsin has an optimum pH of about 1.5. Trypsin, another enzyme of the digestive system, works in the small intestine and its optimum pH is about 8.

Higher or lower pHs than the optimum pH markedly affect the enzyme activity because of the changes in ionization of some amino acid side chains present in the enzyme**.**

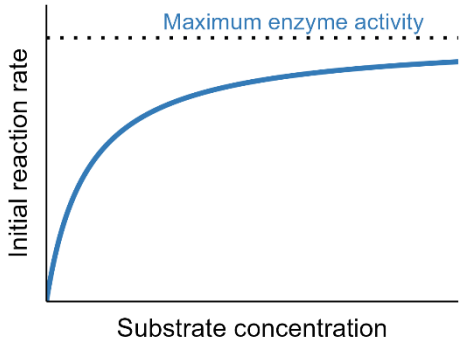


In this example, an increase in pH results in the deprotonation of an amino group located in the active site of the enzyme, which disrupts the interaction with the substrate.

Effect of substrate concentration on the rate of enzyme-catalyzed reactions

As the concentration of the substrate increases, so does the rate of the enzyme-catalyzed reactions. However, the rate of enzyme activity does not increase forever. In most cases, there is a **hyperbolic relationship between the initial rate and the concentration of substrate**:

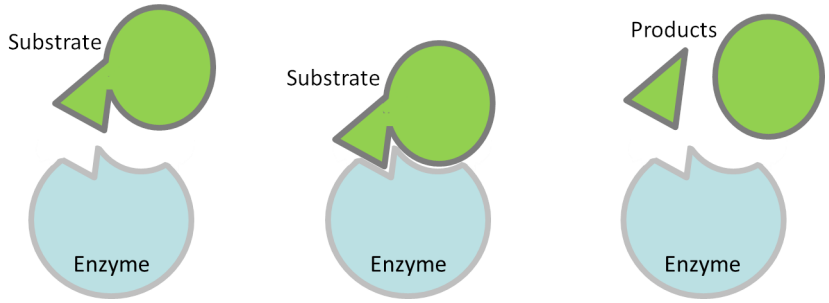
- At low concentration of substrate, there is a steep increase in the rate of reaction with increasing substrate concentration.
- As substrate concentration increases, an increasing proportion of enzyme active sites are occupied. At sufficiently high substrate concentrations, the enzyme becomes saturated with substrate.



When the enzyme is saturated with substrate, further increase in substrate has no effect on rate.

The **Michaelis-Menten model**, proposed by Leonor Michaelis and Maud Menten in 1913, explains this kinetic behavior. The model postulates that the enzyme (E) first combines reversibly with its substrate (S) to form an enzyme-substrate complex (ES). In a subsequent step, the complex breaks down to yield the free enzyme and the reaction product (P):

$$E + S \rightleftarrows ES \rightarrow E + P$$



From this model, an equation is derived expressing the relationship between the concentration of substrate and the initial rate of the reaction:

$$V_0 = \frac{V_{max} \cdot [S]}{K_m + [S]}$$

V_{max} (maximum rate) is the rate of reaction when the enzyme is fully saturated with substrate. The value of V_{max} depends on the **turnover number** of the enzyme (k_{cat}) and the concentration of enzyme:

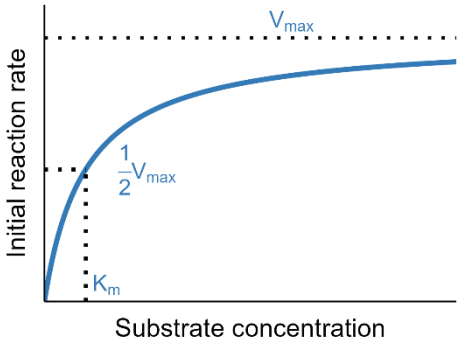
$$V_{max} = k_{cat} \cdot [E]_T$$

(where $[E]_T$ is the total enzyme concentration)

k_{cat} is the maximum number of substrate molecules converted to product per enzyme active site per unit time when the enzyme is saturated with substrate. In other words, it is the number of catalytic cycles per unit of time. In the international system, k_{cat} has units of s^{-1}.

K_m, the Michaelis constant, is often used as an approximate indicator of enzyme-substrate affinity: a lower K_m generally indicates that a lower substrate concentration is required to reach a given reaction rate. However, K_m is not strictly a dissociation constant. K_m has units of molar concentration.

For practical purposes, K_m is equal to the substrate concentration at which the enzyme catalyzes the conversion of substrate molecules at half its maximum rate.



Finally, the ratio k_{cat}/K_m, known as **catalytic efficiency**, reflects how efficiently an enzyme uses its substrate and is particularly useful for comparing enzymes or substrates at low substrate concentrations. Its units are $M^{-1}s^{-1}$.

Exercise: Use this **webtool** to evaluate how changes in kinetic parameters, the enzyme concentration or the substrate concentration affect the reaction rate of an enzyme-catalyzed reaction. Among others, you will be able to observe that:

- V_{max} increases if enzyme concentration increases.
- K_m does not change when enzyme concentration changes.
- A change in K_m indicates a change in the substrate concentration required to reach half V_{max}.
- k_{cat} describes turnover at substrate saturation.
- k_{cat}/K_m is useful for comparing catalytic efficiencies.

Importance of the relationship between the K_m and the physiological concentration of substrate

The relationship between the K_m of an enzyme and the physiological concentration of its substrate permits the prediction of whether or not the rate of formation of product will be affected by the availability of substrate.

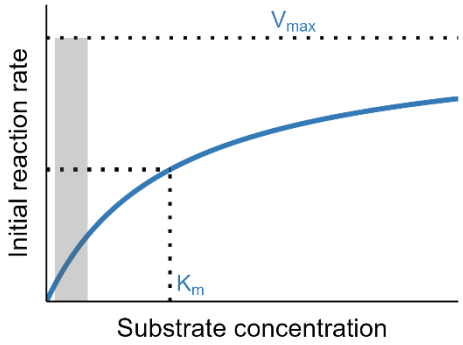
At low substrate concentration as compared to K_m, where $K_m \ggg [S]$, the Michaelis-Menten equation becomes:

$$V_0 = \frac{V_{max} \cdot [S]}{K_m}$$

(because if $K_m \ggg [S]$, it can be assumed that $K_m + [S] \approx K_m$)

This equation corresponds to the equation of a first-order reaction:

$$V_0 = \text{constant} \cdot [S]$$



The activity of an enzyme with a high K_m relative to the physiological concentration of substrate (shadowed area) will be highly dependent on the availability of substrate.

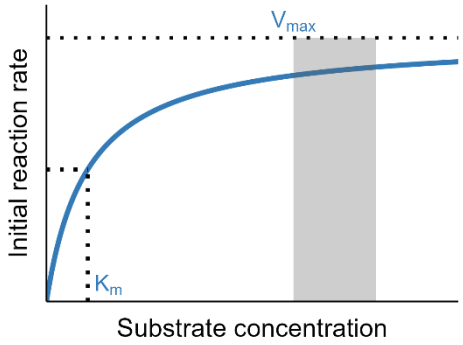
On the other hand, at high substrate concentration, where $K_m \lll [S]$:

$$V_0 = V_{max}$$

(because if $K_m \lll [S]$, it can be assumed that $K_m + [S] \approx [S]$)

This equation corresponds to the equation of a zero-order reaction:

$$V_0 = \text{constant}$$



An enzyme with a low K_m relative to the physiological concentration of substrate (shadowed box) will act at a quite constant rate, regardless of variations in substrate concentration.

How to determine K_m and V_{max}

The values of K_m and V_{max} can be obtained by plotting the initial reaction rate against substrate concentration. However, determining K_m and V_{max} from hyperbolic graphs can be difficult.

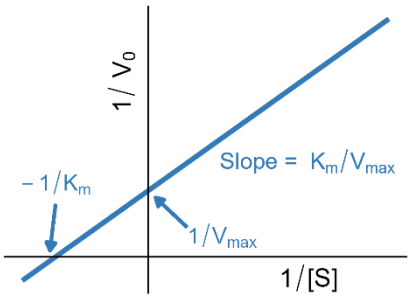
Historically, several linear transformations of the Michaelis-Menten equation have been used to estimate kinetic parameters. One of the best known is the **double-reciprocal plot**, also known as the **Lineweaver-Burk equation**, taking the reciprocal of both sides of the Michaelis-Menten equation:

$$\frac{1}{V_0} = \frac{K_m + [S]}{V_{max} \cdot [S]}$$

In two steps, the equation can be transformed into the following expression:

$$\frac{1}{V_0} = \frac{K_m}{V_{max}} \cdot \frac{1}{[S]} + \frac{1}{V_{max}}$$

This expression corresponds to a linear equation (a = bx + c). Therefore, plotting $1/V_0$ against $1/[S]$ gives a straight line, where the slope is K_m/V_{max}, the y-intercept is $1/V_{max}$ and the x-intercept is $-1/K_m$.



Nowadays, K_m and V_{max} are generally estimated by nonlinear regression of the Michaelis-Menten equation, but the Lineweaver-Burk plot is useful for understanding the relationship between these kinetic parameters and practicing their determination.

Practical determination of kinetic parameters

In this **exercise**, you will learn how to plot the experimental results obtained after the measurement of the rate of an enzyme-catalyzed reaction, and how to calculate graphically the values of K_m and V_{max}.

You will work with a fixed amount of enzyme and will test the initial rate of the reaction after the addition of different amounts of substrate.

The steps of the experiment will be:

- Mix enzyme + substrate.

- Monitor substrate disappearance as a function of time and determine the initial reaction rate (mM/s).
- Plot the reciprocal values of this initial rate and the substrate concentration.
- Change substrate concentration and repeat the previous steps. Do it four more times.
- Fit the data to a straight line and determine the kinetic parameters.

Unfortunately, you will not carry out the experiment with your hands, but still you can imagine it.

The table below contains data that could have been obtained measuring the reaction rate after the addition of five different substrate concentrations (ranging from 0.2 to 20 mM) to a fixed concentration of enzyme (1 µM):

[S] (mM)	V_0 (mM/s)	1/[S]	$1/V_0$
0.2	0.04		
1	0.13		
2	0.19		
4	0.24		
20	0.31		

From these data, you have to obtain the reciprocal values from all substrate concentrations (1/[S]) and rates ($1/V_0$) and plot these values on graph paper.

You should get a straight line. From the intercepts of the line with the axes you will be able to calculate **K_m** and **V_{max}**. Remember that the y-intercept corresponds to $1/V_{max}$ and the x-intercept to $-1/K_m$.

Finally, you can calculate the turnover number of the enzyme (**k_{cat}**) as the ratio between V_{max} and the enzyme concentration (in this case, 1 µM), and the **catalytic efficiency** as the ratio between k_{cat} and K_m. Remember to express V_{max} and enzyme concentration in compatible concentration units.

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